

1-4 Dimensions of Physical Quantities



Source / 來源

- 選修物理(一)課堂講義 (4623R52) 教用版，第1章 不確定度，§ 1-4 物理量的因次，pp. 13–17
- Halliday, Resnick & Walker, *Fundamentals of Physics*, Ch 1: Measurement
- Young & Freedman, *University Physics with Modern Physics*, Ch 1: Units, Physical Quantities, and Vectors

Examples

SI Base Quantities and Dimensions

The seven SI base quantities are mass (M , kilogram), length (L , metre), time (T , second), temperature (Θ , kelvin), electric current (I , ampere), amount of substance (N , mole), and luminous intensity (J , candela). Mechanics mainly uses M , L , and T .

Dimensions are written in square brackets. If a mechanical quantity Q has dimension $M^a L^b T^c$, then $[Q] = M^a L^b T^c$. Two rules are fundamental:

- Both sides of a physical equation must have the same dimension.
- Only quantities with the same dimension may be added or subtracted.

Quantity	SI unit	Dimension
Area	m^2	L^2
Volume	m^3	L^3
Speed	m/s	LT^{-1}
Acceleration	m/s^2	LT^{-2}
Force	$\text{N} = \text{kg} \cdot \text{m/s}^2$	MLT^{-2}
Momentum	$\text{kg} \cdot \text{m/s}$	MLT^{-1}
Work	$\text{J} = \text{kg} \cdot \text{m}^2/\text{s}^2$	ML^2T^{-2}
Kinetic energy	J	
Torque	$\text{N} \cdot \text{m}$	
Angle	rad	1 (dimensionless)
Angular speed	rad/s	T^{-1}
Angular momentum	$\text{m} \cdot \text{kg} \cdot \text{m/s}$	ML^2T^{-1}

Example 1 — Dimension of Planck's Constant (*p. 14*)

The photon-energy relation is $E = hf$, where f is frequency. Find the dimensions of Planck's constant h .

(A) M

(B) ML^2T^{-1}

(C) ML^2T^{-2}

(D) $ML^{-1}T$

(E) MLT^{-2}

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 Solution

Related Problem 1 — Dimension of the Gravitational Constant (*p. 14; 類題 1*)

Newton's law of gravitation is $F = GMm/r^2$. Find $[G]$.

(A) $L^{-1}M^{-3}T^{-2}$

(B) $L^{-1}M^{-2}T^3$

(C) $L^3M^{-1}T^{-2}$

(D) $L^3M^{-2}T^{-3}$

(E) $L^{-3}MT^2$

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Example 2 — Dimensional Analysis of Momentum (*p. 14*)

Newton's general law is $F = \Delta p / \Delta t$. Suppose $p = m^x v^y$. Find x and y .

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Related Problem 2 — Free-Fall Speed (*p. 14; 類題 2*)

For an object dropped from rest through a distance h , suppose $v = cm^x h^y g^z$, where c is dimensionless. Find (x, y, z) .

(A) $(1, \frac{1}{2}, -\frac{3}{2})$

(B) $(0, \frac{1}{2}, -\frac{1}{2})$

(C) $(\frac{1}{2}, 1, -\frac{1}{2})$

(D) $(0, \frac{1}{2}, \frac{1}{2})$

(E) $(\frac{3}{2}, -\frac{1}{2}, \frac{1}{2})$

 Solution

Quick Check

Q1 — Dimension of Pressure

Pressure is $P = F/A$. Which dimension is correct?

(A) MLT^{-2}

(B) MLT^{-1}

(C) ML^2T^{-2}

(D) $ML^{-1}T^{-2}$

(E) ML^3

 Solution

Q2 — Dimension of a Low-Speed Drag Coefficient

At low speed, air resistance is $F = cv$. What is the dimension of c ?

(A) LT^{-1}

(B) L^2MT

(C) LM^2T^{-1}

(D) MT^{-1}

(E) T^{-1}

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 Solution

Q3 — Complete the SI Units and Dimensions

Complete the table.

Quantity	SI unit	Dimension
Density D		
Speed v		
Work W		
Momentum p		
Force F		

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 Solution

Challenge

Q1 — Sound Speed in a One-Dimensional Atomic Model

In a simplified one-dimensional metal model, atoms are masses m separated by distance d and joined by springs of elastic constant k . Suppose $v = Cm^ad^bk^c$, where C is dimensionless. Find a , b , and c , then write v in terms of m , d , and k .

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Q2 — How Many Quantities Share a Dimension?

Among momentum, force, kinetic energy, heat, and work, how many have the same dimension?

(A) 1

(B) 2

(C) 3

(D) 4

(E) 5

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Exam Archive

Q1 — Water-Wave Speed by Dimensions (2021, 110 指考)

In a ripple tank, wave speed may depend on gravitational acceleration g , water density ρ , and depth D . Using dimensions only, which proportionality can have the dimension of

speed?

(A) $v \propto gD$

(B) $v \propto \rho gD$

(C) $v \propto \sqrt{gD}$

(D) $v \propto g\sqrt{\rho D}$

(E) $v \propto 1/\sqrt{gD}$

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 Solution

Q2 — Dimension Matching in Periodic Motion (2020, 109 指考)

A particle executes periodic motion. The mean square displacement is X^2 , the mean square momentum is P^2 , and the total mechanical energy is E . Which printed expression is intended to have the dimension of a period?

(A) $\sqrt{X^2 P^2 / E}$

(B) $X^2 P^2 / E$

(C) $\sqrt{X^2 E^2 / P^2}$

(D) $E \sqrt{X^2/P^2}$

(E) $E \sqrt{P^2/X^2}$

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 Solution

Glossary

English term	中文	Definition / Note
SI base quantity	SI 基本量	One of the seven quantities defining the SI system
dimension	因次	Representation such as $M^a L^b T^c$
dimensional analysis	因次分析	Checking or deriving relations using dimensions
mass	質量	Base quantity with dimension M
length	長度	Base quantity with dimension L
time	時間	Base quantity with dimension T
force	力	$F = ma$; dimension MLT^{-2}
momentum	動量	$p = mv$; dimension MLT^{-1}
work	功	Force through a displacement; dimension ML^2T^{-2}
kinetic energy	動能	Energy of motion; dimension ML^2T^{-2}
potential energy	位能	Energy associated with position or configuration
power	功率	Rate of doing work
pressure	壓力	Force per area; dimension $ML^{-1}T^{-2}$
torque	力矩	Force times lever arm; dimension ML^2T^{-2}
angular momentum	角動量	Length times momentum; dimension ML^2T^{-1}
spring constant	彈性常數	Force per extension; dimension MT^{-2}
dimensionless	無因次	Has dimension 1, as for radians